RISHI GARG

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EDUCATION

B.TECH(Software Engineering)	2025-2029	Delhi Technological University, New Delhi	**CGPA: 9.70**
CBSE (Class XII)	2025	Yuvashakti Model School	**96.4 %**
CBSE (Class X)	2023	Yuvashakti Model School	**95.4 %**

WORK EXPERIENCE

AI/ML Researcher | Vision and Interpretability Lab (VAIL) **August 2026-Present**

- Researching on computer vision and multimodal representation learning. **Top 20** finish, Track 2 of the NeurIPS 2026 RealPDE Competition with a final score of **81.82**, competing against leading teams worldwide.

AI/ML Research Internship | Delhi Technological University **June 2026 - Present**

- Conducting research on complexity-reduction strategies for the **ChARM architecture**, with emphasis on scalable temporal and inter-channel modeling in multimodal time-series learning.

ACADEMIC PROJECT

Andrew Ng Digital Twin - Persistent Scientist AI System | GitHub | Live Demo | Video Demo | **July 2026**

- Deployed a role-adaptive Andrew Ng digital twin grounded in **529** sources, delivering citation-backed multi-turn conversations with **21/21 completed answers** passing deterministic persona and response-structure checks.
- Built timeline-aware graph memory, both in and cross-session recall, a live memory dashboard and streaming synthetic cloned voice; validated **93.3% of 15** memory scenarios, **95.2% of 21** assertions.

Octave - Context-Aware Spatial Interface | GitHub | VideoDemo | **May 2026**

- Developed a multimodal Python/C++ interface combining offline NLP+CV**,** engineering dynamic context routing to translate **8** base gesture families into **15+** actions with **<5ms IPC** and **<80ms** end-to-end latency**.**
- Engineered a React/Electron dashboard for live retraining via **Transfer Learning** without catastrophic forgetting, hot-swapping PyTorch LSTMs/FFNNs (**~99%** static on **2,100 samples**, **~95.8%** dynamic) with **zero downtime**.

Bandhu.AI - Voice-Driven Universal Digital Inclusion System | GitHub | Video Demo | **April 2026**

- Architected an interactive voice-to-web agent bridging **100M+ rural citizens** to government portals, processing DTMF and **22+ dialects** with **<3s response** and **<150ms API** latency to map unstructured audio into stateful JSON.
- Engineered an autonomous backend pipeline (FastAPI, Playwright) parsing JSON to submit **20+ dynamic** form fields in **<45 seconds**, reducing registration time from a **3-hour physical process to a 5-minute automated call**.

Gesture-Controlled Drone System | GitHub | Video Demo | **January 2026**

- Developed an edge-optimized, multimodal drone control interface fusing gesture, voice, and facial recognition (smile-based speed modulation), processing **7 distinct** hand gestures at **~30 FPS**.
- Implemented temporal smoothing to eliminate prediction jitter, achieving **~36ms end-to-end** latency and **~93% accuracy** via a custom **ANN/CNN** architecture.

ACADEMIC ACHIEVEMENTS AND AWARDS

- **Academic Rank: 2nd** in Software Engineering Department **(9.7 CGPA).**
- **Top 20 Participant (Track 2):** Internationally in **NeurIPS 2026 RealPDE Competition**, final score **81.82.**
- **Top 10 Participant (PS11):** Nationally in **ISRO Bhartiya Antariksh Hackathon 2026** for making PT-JEPA.
- **Top 30 Participant:** National-Level **SahAI for Shiksha Hackathon 2026** by **Wadhwani AI** & **Google.org.**
- **Top 10 Participant:** **KRITI Social Impact Challenge '26** at BITS Pilani (APOGEE 2026).
- **JEE Main 2025** : **99+ Percentile(Top 1%)** nationwide (**AIR 14182 among 1M+ candidates**).

TECHNICAL SKILLS

- **Languages**: C, C++, Python, SQL
- **AI/ML:** PyTorch, Scikit-learn, Statsmodels, OpenCV, MediaPipe, Pandas, NumPy, RAG, STT, Chatterbox TTS.
- **Tools & Frameworks**: Git/GitHub, FastAPI, React, Hugging Face, Playwright, Electron, Docker, PyPI, Vercel.
- **Certificates**: **DL Specialization, Sequence Models, Convolutional Neural Networks, ML Specialization, Basics of Quantum Information**

POSITIONS OF RESPONSIBILITY

- **Co-Head, AIMS-DTU**